

Geometry-first design of organic/inorganic multilayer thin-film encapsulation via a literature-calibrated digital twin: defect-mediated moisture physics, zero-refit validation, and confidence-qualified design rules

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Highlights. Defect-mediated 1-D twin predicts multilayer WVTR to $+0.11/-0.01$ decade, zero refit. · Transient lag phase explains sub-detection-limit dyad data; consistent under censoring. · Exhaustive enumeration provides ground truth; NSGA-II verified for scale-up (GD 0.06). · Number-verified causal chains from exact Shapley/Sobol. · Robust rules: 22.5 [21–26]–44 [38–48] nm inorganic window; linear pair law; device-anchored M_{crit} floor.

Abstract

Multilayer thin-film encapsulation (TFE) gates the deployment of moisture-sensitive optoelectronics, including perovskite photovoltaics, yet its geometric design space—organic/inorganic thicknesses and dyad number—is sampled experimentally at only a few points. We report a lightweight digital twin that treats TFE design as a geometry problem: one-dimensional moisture transport in water-activity form with defect-mediated effective media (nucleation-driven pinhole closure, a particle floor, a cracking penalty, and lateral tortuous-path decoupling), coupled heat transfer, and an exact layer-lumped time lag; the transport core is verified against the analytic Fickian solution (0.45% maximum error). Demonstrated end-to-end on the ALD- Al_2O_3 /parylene-C literature system and calibrated only on a published single-layer thickness series plus the one-dyad point of the same source, the model predicts the two- and three-dyad WVTR with zero refitting to within $+0.11$ and -0.01 decades once a finite multi-day run is simulated (insensitive across 2.0–5.5 days; steady state probably excluded)—identifying the measurements as lag-phase-limited and remaining consistent with their below-detection (censored) status. The inferred pinhole density (8.5 mm^{-2}) lies within independent Ca-test ranges. Exhaustive enumeration of 4,104 designs supplies a ground-truth six-objective Pareto front, against which a dependency-free NSGA-II—retained as verified machinery for higher-dimensional extensions—is benchmarked (generational distance 0.060); device temperature and transmittance vary by only 0.10 K and 0.35 percentage points, collapsing the problem to four effective objectives. Exact interventional Shapley values and Sobol indices on the factorial grid are translated into causal chains carrying model-internal quantities; this number-embedding convention exposed and corrected an over-interpretation of the organic-thickness effect. Monte-Carlo uncertainty propagated through the entire calibration pipeline ($N=500$) yields confidence-qualified rules: an inorganic window of 22.5 [21.0–26.0]–44.0 [37.5–48.0] nm bounded by pinhole closure and cracking, and a pair law linear in all draws ($R^2 \geq 0.9998$). A first device-anchored calibration of the critical moisture dose ($M_{\text{crit}}^{\text{eff}} \geq 0.97 \text{ g m}^{-2}$, from published encapsulated-device aging) implies moisture-limited lifetimes of ≥ 14 years for a single dyad even at the window edge (~ 26 at the 42 nm optimum) and ≥ 99 years

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window-wide for $n \geq 2$ —moisture ceases to be the binding lifetime channel for dyad stacks—while all thresholds and rankings are unchanged. Code, database and provenance are released.

Keywords: thin-film encapsulation; water vapor transmission rate; atomic layer deposition; pinhole defects; multi-objective optimization; uncertainty quantification

1 Introduction

Perovskite photovoltaics now rival silicon in efficiency, but moisture sensitivity makes encapsulation a decisive step toward deployment; recent reviews stress the dual requirement of blocking water ingress and lead egress and the absence of standardized stability protocols [1, 2]. Among encapsulation routes, multilayer thin-film encapsulation—alternating dense inorganic barriers with organic interlayers—offers conformality, flexibility and a low thermal budget compatible with perovskite devices [3–7].

Experiment has established the ingredients. Low-temperature ALD Al_2O_3 single layers protect devices but plateau in barrier performance [4–6], consistent with the ALD-barrier benchmarks of Carcia *et al.* [19]; organic/inorganic dyads improve WVTR by orders of magnitude, with a 3-dyad pV3D3/ Al_2O_3 stack near $10^{-4} \text{ g m}^{-2} \text{ day}^{-1}$ at $38^\circ\text{C}/90\% \text{ RH}$ [3] and a 3-dyad parylene-C/ Al_2O_3 stack below instrument detection [7]; ALD nanolaminates extend the same logic within the inorganic phase [20]; and residual particles introduced during deposition dominate the defect budget of practical barriers [7]—particles whose burial is a principal *planarization* function of the organic layers [21]. Yet these studies sample the geometry (d_{org} , d_{in} , n) at a handful of points each, and the recent multi-property figure of merit for TFE [15] confirms that barrier, optical and mechanical demands must be balanced while still collapsing them into a single scalar ranking of fabricated films.

The transport physics needed to map this space is long established. Permeation through coated polymers is defect-dominated: single-pinhole models [17] and defect–permeation correlations on coated PET [18] quantified it two decades ago; Graff *et al.* showed that multilayer performance is governed by the *lag time* of long lateral paths between offset defects rather than by equilibrium permeation [8]; ideal-laminate models reproduce time-dependent multilayer permeation [9]; and moisture ingress is modeled at module scale [10]. These tools, however, have served as isothermal, mostly steady-state analyses of fabricated structures—not as forward engines coupled to environmental dynamics, durability, lifetime and systematic design-space search.

Machine learning has entered adjacent problems from another direction: random-forest/SHAP identification of stabilizing capping-layer chemistry [11], NSGA-II thickness optimization for optoelectronic objectives [12, 13], and an AutoML processing pipeline described as an early perovskite digital-twin component [14]. None targets encapsulation geometry, couples a transport PDE to the search, treats reliability-centred objectives, or grounds attribution in mechanism—SHAP answers *what mattered*, not *why*.

Here we assemble these validated pieces on a deliberately scoped claim: a *barrier-stack* digital twin, demonstrated and validated end-to-end on the Al_2O_3 /parylene-C system of Ref. [7], *toward* perovskite encapsulation, whose device-specific transfer requirement—the critical moisture dose—we identify and, in §3.4, floor with published device data. Contributions: (i) a ≈ 1 s-per-design defect-mediated 1-D moisture/heat twin whose geometry sensitivity derives from explicit closure, particle-floor, cracking and tortuous-path physics [8, 17, 18]; (ii) a disclosed calibration/validation protocol—shape from a

single-layer series, scale from one multilayer point, zero-refit prediction of the rest—including a transient, censoring-consistent re-reading of sub-detection measurements; (iii) a six-objective Pareto exploration verified against exhaustive enumeration, with exact Shapley/Sobol attribution rendered as causal chains carrying model-internal numbers; and (iv) design rules qualified by Monte-Carlo uncertainty propagated through the calibration pipeline itself. We claim no new material, transport law or algorithm; the contribution is the validated integration and the design map and rules it produces.

2 Model and methods

2.1 Governing model

Moisture transport is solved in water-activity form, $S(x) \partial_t a = \partial_x(P(x, T) \partial_x a)$, with $a = C/S$ continuous across interfaces and equal to RH at the ambient boundary; sorption is activity-linear (Henry regime; revisited in §3.5). The device boundary is a perfect sink ($a=0$), matching Ca/MOCON geometry [7, 17]; for barriers on devices, vapor accumulation beneath pinholes reduces the driving force [17], so the sink assumption is *conservative*. Geometry sensitivity is carried by defect-mediated effective media: $P_{\text{in}}^{\text{eff}} = P_{\text{lat}}(T) + f(d) P_{\text{org}}(T)$ with

$$f(d) = f_0 e^{-d/d_c} + f_{\text{res}} + f_{c0} \left[\frac{d - d_{\text{crit}}}{d_{\text{crit}}} \right]_+^2, \quad (1)$$

encoding nucleation-driven closure, a thickness-independent particle floor [7, 18], and channel cracking; sandwiched organic layers carry $P_{\text{org}}^{\text{eff}} = P_{\text{org}}/\tau^2$,

$$\tau^2 = 1 + \frac{s^2 \ln(s/r)}{2\pi d_{\text{org}}^2}, \quad s = r\sqrt{\pi/f}, \quad (2)$$

the 1-D reduction of the lateral detour between offset pinholes [8], valid for $s \gg d_{\text{org}}$ (here $s = 343\text{--}459 \mu\text{m}$ against $d_{\text{org}} \leq 1 \mu\text{m}$; derivation SIS3). Heat conduction with solar source and convective/sky-radiative boundaries supplies $T(t)$ for Arrhenius coupling and the cycling amplitude of a Basquin fatigue proxy with $\sigma_c(d) = \sigma_{c0} \sqrt{d_{\text{ref}}/d}$. The moisture-limited lifetime is a dose-threshold proxy,

$$t_{80}^{\text{dose}} = t_{\text{lag}} + M_{\text{crit}} R_{\text{tot}}/\Delta a \quad (3)$$

(written T80 in figures; it maps onto the photovoltaic T80 only after device-level dose calibration), with the exact layer-lumped Frisch lag [22] $t_{\text{lag}} = \sum_j C_j R_{L,j} R_{R,j} / R_{\text{tot}}$ (reducing to $L^2/6D$; SIS1). The finite-volume, harmonic-conductance, implicit solver reproduces the analytic Fickian breakthrough flux to 0.45% (Fig. 2).

2.2 Literature database and provenance

All inputs live in a relational database with per-row sources, conditions, censoring flags ($=, <, \sim$), quality grades, and an enforced calibration/validation role split. Graphical data were digitized by automated marker detection with archived axis-calibration overlays (residuals ≤ 0.14 decade; ± 0.15 decade per point).

2.3 Calibration and zero-refit validation

Stage A fits the *shape* of $f(d)$ to the single-layer Al_2O_3 series of Ref. [7] (15–60 nm, 38 °C/100% RH; max residual 0.047 decade) under the a-posteriori-consistent assumption $\text{WVTR}_{\text{bare}} \propto f(d)$ (bare-pinhole conductance 5.8×10^{-6} of the open-vapor limit, i.e., access-limited [17, 18]). The falling branch is constrained by one datum, leaving d_c only *upper-bounded* (≤ 1.8 nm at the measurement-noise level, f_0 compensating; profile analysis SIS3)—closure completes below the first measured thickness—so no statement is extrapolated below 15 nm. Stage B sets f_{res} from the one-dyad point of the same source (50 nm Al_2O_3 /500 nm parylene-C dyads; the anchor thickness was corrected from an initial 30 nm misreading upon full-text verification, with every downstream quantity regenerated—all shifts fall inside the reported CIs). The two- and three-dyad measurements are then *predicted with zero refitting*—at steady state and as apparent WVTR after a finite multi-day run (the article specifies the coulometric detection limit, $5 \times 10^{-5} \text{ g m}^{-2} \text{ day}^{-1}$, but not the run duration; predictions are insensitive across 2.0–5.5-day durations, SI Fig. S3, and steady state is excluded regardless, SIS2); because both points lie below the stated steady-state detection limit they are treated as *censored*, and validation is assessed as consistency under censoring plus descriptive decade errors. The article’s separate 3-dyad result at 30 nm (its Fig. 4, $< 10^{-5}$, detection-limited) is likewise reproduced as a lag-phase reading (corrected-model steady state $3.5 \times 10^{-5} \text{ g m}^{-2} \text{ day}^{-1}$). The within-source split is disclosed; complementary checks are an independent cross-temperature point (50 nm at 45 °C [4]), the collapsed-benefit regime of particle-rich films [7], and the solubility consistency of §3.1. Falsification targets: the 1–4-dyad WVTR *and lag* series of Ref. [3] (Fig. 2d) and any ≤ 10 nm ALD point.

2.4 Six-objective exploration

Design variables: $d_{\text{org}} \in [100, 1000]$ nm (planarization/process bound; §3.5), $d_{\text{in}} \in [15, 120]$ nm, $n \in \{1 \dots 6\}$. Objectives: t_{80}^{dose} at 38 °C/90% RH; peak device temperature under a 15–42 °C diurnal cycle (1000 W m^{-2} peak; organic-capped stacks, hence fixed top emissivity); visible transmittance; a *relative* cost proxy equal to measured deposition time on one toolset (ALD 0.098 vs. parylene 6 nm min^{−1} [7]); areal mass; and the fatigue proxy. The analytic evaluator equals the PDE solver at steady state to machine precision and is lag-exact, enabling exhaustive enumeration of 4,104 designs as ground truth. Because three variables permit enumeration, NSGA-II is *not* claimed as a contribution; a dependency-free implementation (pop. 80, 80 generations, SBX/polynomial) is benchmarked against the exhaustive front as verified machinery for higher-dimensional extensions where enumeration fails.

2.5 Exact explainability and number-verified chains

On the full-factorial grid, interventional Shapley values follow exactly from the 2^3 coalition group means under the uniform grid measure (efficiency identity $\leq 10^{-11}$ of the response), and Sobol shares from exact functional ANOVA under the same measure—no surrogate or sampling approximation (elementary with three features; the point is exactness where available). Each high-importance driver→objective pair is rendered as a causal chain whose arrows carry model-internal quantities ($f, s, \tau^2, R_{\text{in}}, R_{\text{sand}}, t_{\text{lag}}, \sigma_c/\sigma_a$) recomputed from probes, with a regression check of the claimed law.

2.6 Uncertainty quantification through the calibration pipeline

$N=500$ Monte-Carlo draws perturb measured inputs (single-layer series $\sigma=0.08$ decade; digitized anchor 0.10) and placeholder properties (parylene $D \times / \div 3$; S within the validation-constrained $0.5\text{--}2\times$; E_a $30\text{--}50 \text{ kJ mol}^{-1}$; pinhole radius $\times / \div 2$; $M_{\text{crit}} \times / \div 3$; $\sigma_{c0} \times / \div 1.5$; fatigue exponent $6\text{--}10$; rationale SI Table S4), re-running the *entire* Stage-A/B calibration per draw so calibration-induced correlations are preserved. Outputs separate cleanly: regime thresholds are measurement-noise-dominated, whereas absolute lifetime and durability feasibility are prior-dominated (§3.4).

3 Results and discussion

3.1 Calibrated pinhole physics and a transient, censoring-consistent reading

The fitted $f(d)$ (Fig. 3a) reproduces the non-monotonic single-layer series—closure to a defect floor by $\sim 20 \text{ nm}$, degradation beyond $\sim 50 \text{ nm}$ [7]—and implies $8.5 \text{ pinholes mm}^{-2}$ ($343 \mu\text{m}$ spacing), within Ca-test ranges [18, 19]: a plausibility check independent of the fit. Zero-refit *steady-state* prediction of the dyad series fails structurally: the measured improvement is geometric ($\div 4.7$ per dyad, $R_3/R_1 \approx 22$ required), whereas the ladder obeys

$$\frac{R_{\text{sand}}}{R_{\text{in}}} = \frac{d_{\text{org}}}{d_{\text{in}}} \left[f + \frac{r^2 \ln(s/r)}{2d_{\text{org}}^2} \right], \quad (4)$$

bounding $R_3/R_1 \leq 7.8$ for all admissible parameters (SI S2)—the $+0.75$ -decade 3-dyad overshoot is a model-class statement, not a tuning failure. Simulating the finite $\approx 72 \text{ h}$ run resolves it (Fig. 3b,c): apparent WVTR matches the 2-dyad point to $+0.11$ decade and the 3-dyad point to -0.01 decade (lag times 67 and 180 h), remains below the stated instrument limit exactly as the censored data do, and stays within ≤ 0.4 decade for durations of 2.0–5.5 days. This identifies the multilayer measurements as lag-phase-limited and reconciles apparent geometric scaling with series-resistance physics [8]. The parylene solubility required for joint agreement ($0.5\text{--}2 \text{ kg m}^{-3}$) coincides with the independent datasheet uptake ($<0.1 \text{ wt\%}$), a non-circular consistency loop. An ablation study (SI S7, Fig. S4) confirms each mechanism is pinned by a distinct data feature: removing any shape term breaks the five-point calibration (≥ 0.43 vs. 0.047 decade max residual), removing the tortuous path or the lag breaks the zero-refit validation (max error 0.33 and 0.69 decade), and the uncalibrated model misses the anchor by 0.70 decade with incorrect n -scaling.

3.2 Pareto structure and effective dimensionality

Across 4,104 designs the peak device temperature spans 0.10 K and transmittance 0.35 percentage points (organic-capped stacks: $\sim \text{mK}$ conduction resistance against $\sim 10 \text{ W m}^{-2} \text{ K}^{-1}$ boundary losses; parylene $n=1.639$ index-matched to Al_2O_3 $n=1.65$). The six-objective problem collapses to four effective objectives (247 designs on the reduced true front)—itself a design insight: thermal and optical performance are not geometric degrees of freedom in this material pair. NSGA-II lands on the exhaustive front (GD 0.060 mean; the larger inverted GD, 0.168, reflects an 80-point archive against a 3,064-point six-objective front—cardinality, not convergence—and is moot in 3-D). Lifetime climbs the dyad ladder linearly ($+8,750\text{--}8,980 \text{ h}$ per pair at baseline) with no in-model saturation; representative designs in Table 2.

3.3 Number-verified causal chains, and a correction they forced

Exact Sobol shares (Fig. 5a) rank d_{in} first for lifetime (0.551), then the $d_{\text{in}} \times n$ interaction (0.184)—structural, since the dominant resistance is the product $nR_{\text{in}}(d_{\text{in}})$, and resonant with the collapsed dyad benefit of particle-rich films [7]: poor barriers cannot be stacked into good ones. Durability loads entirely on d_{in} (1.00) through $\sigma_c \propto d^{-1/2}$, giving $N_f \propto d^{-m/2}$ (log–log slope -3.6 under the fatigue prior). The chain convention proved more than presentational: embedding internals into the $n=1$ organic-thickness chain exposed that R_{top} lies seven orders below R_{in} , so d_{org} is inert for single dyads; our earlier reading of the optimizer’s boundary choice was a tie-breaking artifact, corrected in the published chain (Figs. 5c, 6). The organic layer contributes to lifetime *only* as the lateral decoupling medium of multilayers ($R_{\text{sand}} \propto 1/d_{\text{org}}$ for $n \geq 2$).

3.4 Robust design rules; a device-anchored floor for the dose threshold

Propagating all inputs through re-calibration (Fig. 7): **(G1)** $d_{\text{in}} \geq 22.5$ [21.0–26.0] nm (closure); **(G2)** < 44.0 [37.5–48.0] nm (cracking onset; beyond it lifetime and durability collapse jointly)—both thresholds measurement-noise-dominated; **(G3)** within the window $N_f \propto d^{-m/2}$; a 10-year fatigue target needs $d_{\text{in}} \leq 30$ [16–49] nm and is unreachable at any thickness in 16% of draws under the fatigue prior—an explicit quantification of the proxy’s weakness; **(G4)** the pair law is linear in every draw ($R^2 \geq 0.9998$) with relative gain 0.5 [0.4–0.5] of the 3-dyad lifetime per added dyad, while the absolute gain (8,458 [1,404–50,128] h) is M_{crit} -prior-dominated; **(G5, transport-only, conditional)** do not exceed the planarization-required organic thickness ($R_{\text{sand}} \propto 1/d_{\text{org}}$ for $n \geq 2$; planarization unmodeled [21]); **(G6)** thermal/optical objectives delegated to materials selection; **(G7)** thresholds and rankings invariant under $M_{\text{crit}} \times / \div 3$.

The remaining absolute-scale gap is now floored by a first device-anchored calibration (SI S4.4b). Transferring Ref. [4]’s own measured barrier WVTR to its shelf conditions (vapor-access mechanism; Arrhenius alternative within 15%) and integrating over the reported 7,500 h with $<4\%$ PCE loss yields, within the same sink model, $M_{\text{crit}}^{\text{eff}} \geq 0.97$ [0.90–1.04] g m $^{-2}$ under monotone dose–damage only (a linear reading gives ≥ 4.9). Consequently every design inside the G1–G2 window exceeds ~ 25 years of moisture-limited life even at a single dyad (Table 2, right column): *moisture ceases to be the binding lifetime channel for dyad stacks*, shifting the design problem to durability, cost and weight, while all thresholds and rankings—per G7—are unchanged.

3.5 Limitations and transferability

(i) The d_{org} optimum sits on the imposed 100 nm bound because pinhole replication/planarization through thin organics is unmodeled [21]. (ii) t_{80}^{dose} is a dose proxy; the device-anchored bound of §3.4 is a same-model floor whose transfer assumes the sink bias is non-increasing with barrier resistance (SI S4.4b); a two-sided estimate requires aging beyond threshold. (iii) The cracking onset was calibrated under flexible-substrate handling strain [7]; rigid deployments shift d_{crit} upward. (iv) Durability is a Basquin proxy pending film-mechanics primaries. (v) Calibration and quantitative validation share one source, mitigated by cross-regime prediction, the cross-temperature check [4], solubility consistency, and named falsification targets [3]. (vi) Lifetime ranking is evaluated at 38 °C/90% RH; Arrhenius extrapolation rescales absolute values without reordering geometry. (vii) Activity-linear sorption may break near saturation. (viii) The cost proxy is relative deposition time on one toolset. (ix) The device-side sink is conservative [17].

4 Conclusions

Treating thin-film encapsulation as a geometry problem, a literature-calibrated defect-mediated digital twin—verified against analytic transport; validated zero-refit against multilayer experiments through a transient, censoring-consistent re-reading of sub-detection data; explored to an exhaustively verified Pareto front; explained through exact, number-verified causal chains that corrected our own over-interpretation; and stress-tested by calibration-through uncertainty propagation—yields confidence-qualified design rules: a 22.5–44 nm inorganic window, a saturation-free linear pair law, and a device-anchored dose floor under which moisture ceases to be the binding lifetime channel for dyad stacks. The framework is lightweight and released with database and provenance; its consequential open items—a two-sided perovskite dose threshold and film-mechanics primaries—slot directly into the released pipeline.

Declarations

Data and code, with per-row provenance, censoring and quality flags and digitization overlays, are available at <https://github.com/teoyang0704-wq/Perovskite-TFE-DigitalTwin>; an interactive deployment of the calibrated twin (design sliders, Pareto map, and both M_{crit} tiers) is available at <https://tfe-twin.streamlit.app>. **AI-use disclosure:** portions of code, analysis and text were developed with a large-language-model assistant (Anthropic Claude); all methods, results and claims were verified by the author, who takes full responsibility. CRediT / competing interests / funding: [to complete].

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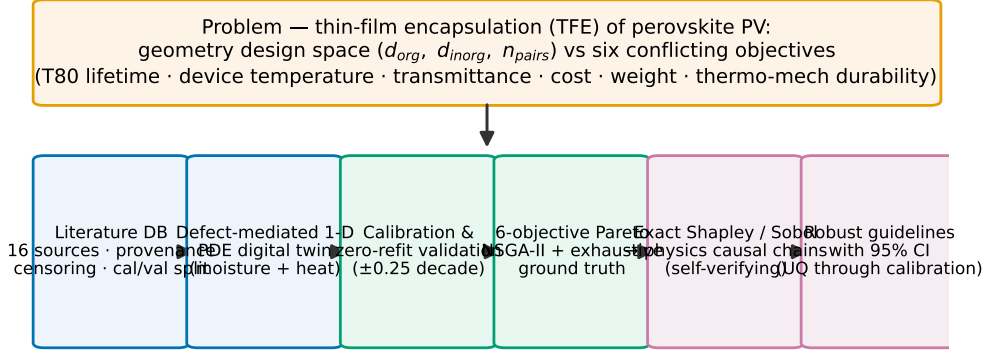


Fig. 1 | Physics-informed digital-twin framework: every arrow is a validated, number-carrying step (Steps 1–8 of this work).

Figure 1: Framework overview: literature database with provenance → defect-mediated 1-D moisture/heat digital twin → calibration and zero-refit validation → six-objective Pareto exploration with exhaustive ground truth → exact Shapley/Sobol translated into number-verified causal chains → robust guidelines from calibration-through uncertainty quantification. Right column: the verified quantity carried by each step.

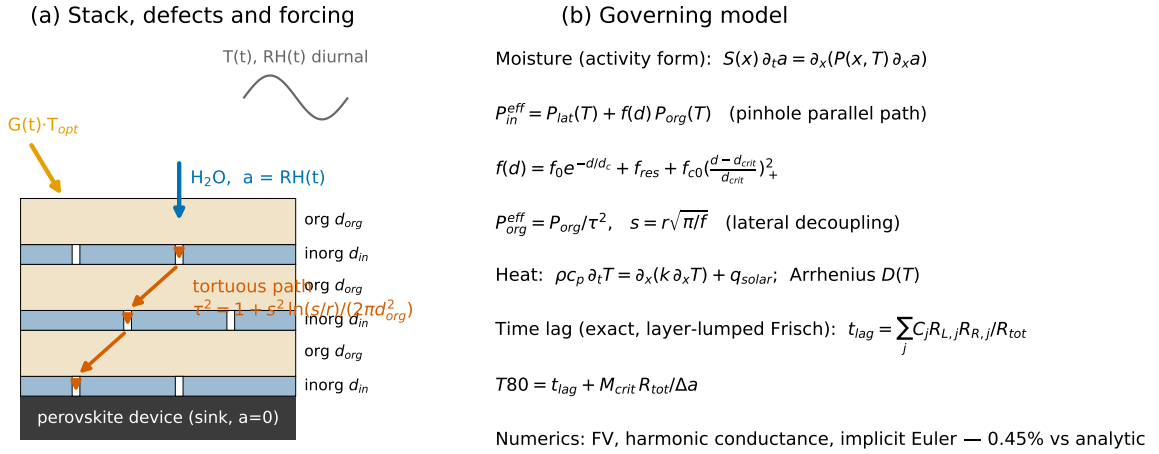


Figure 2: Digital-twin structure. (a) Stack cross-section with laterally offset pinholes and the lateral (tortuous) transport path; environmental forcing (solar flux, diurnal T/RH , ambient water activity). (b) Governing model: activity-form diffusion, defect-mediated effective media $f(d)$ and τ^2 , heat equation, exact layer-lumped Frisch lag, dose-based lifetime, and numerics (0.45% verification).

Table 1: Calibrated quantities (details and priors: SI Tables S1–S4).

Quantity	Value	Provenance
$f(50 \text{ nm})$	6.67×10^{-8} (8.5 nm^{-2} ; $343 \mu\text{m}$)	Stage B anchor (50 nm dyads)
Closure threshold	22.5 [21.0–26.0] nm	UQ ($N=500$)
Cracking onset d_{crit}	44.0 [37.5–48.0] nm	Stage A + UQ
d_c (closure length)	$\leq 1.8 \text{ nm}$ (upper bound)	profile analysis (SI S3)
Identified invariant	$f(15)/f_{res} = 9.1$	SI S3
Pair law (relative)	0.5 [0.4–0.5] per dyad; $R^2 \geq 0.9998$	UQ
M_{crit}^{eff}	≥ 0.97 [0.90–1.04] g m^{-2} (Tier-1)	device-anchored (SI S4.4b)

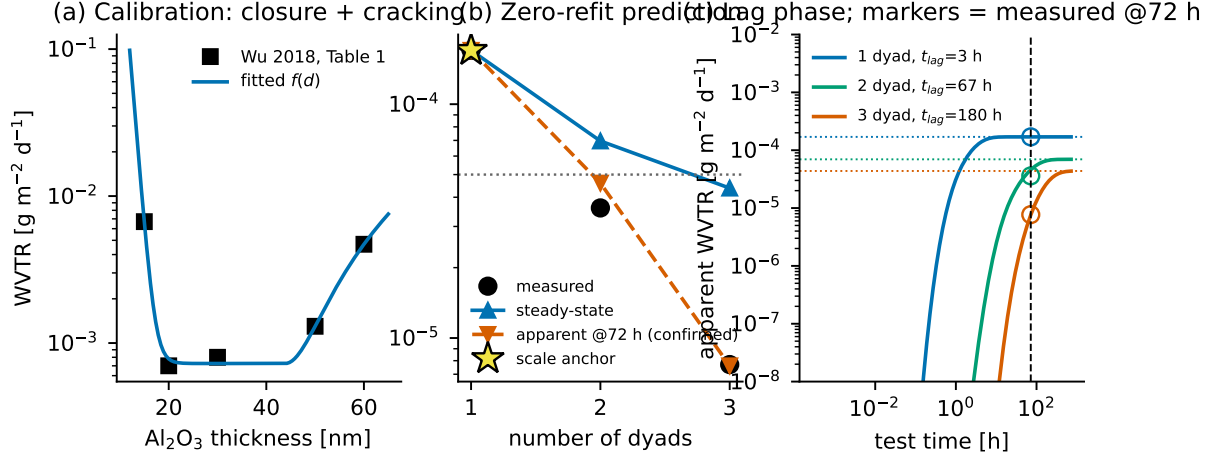


Figure 3: Calibration and zero-refit validation against Ref. [7] (38 °C/100% RH). (a) Single-layer Al₂O₃ thickness series and fitted $f(d)$ (closure + cracking). (b) Dyad series: measurement, steady-state prediction, and apparent WVTR after a finite multi-day run (duration unstated in the source; predictions insensitive across 2.0–5.5 days, SI Fig. S3); star: scale anchor; the 2- and 3-dyad points are censored (below the stated 5×10^{-5} limit). (c) Transient apparent WVTR with lag times; circles: measurements placed at 72 h; dotted: stated instrument limit.

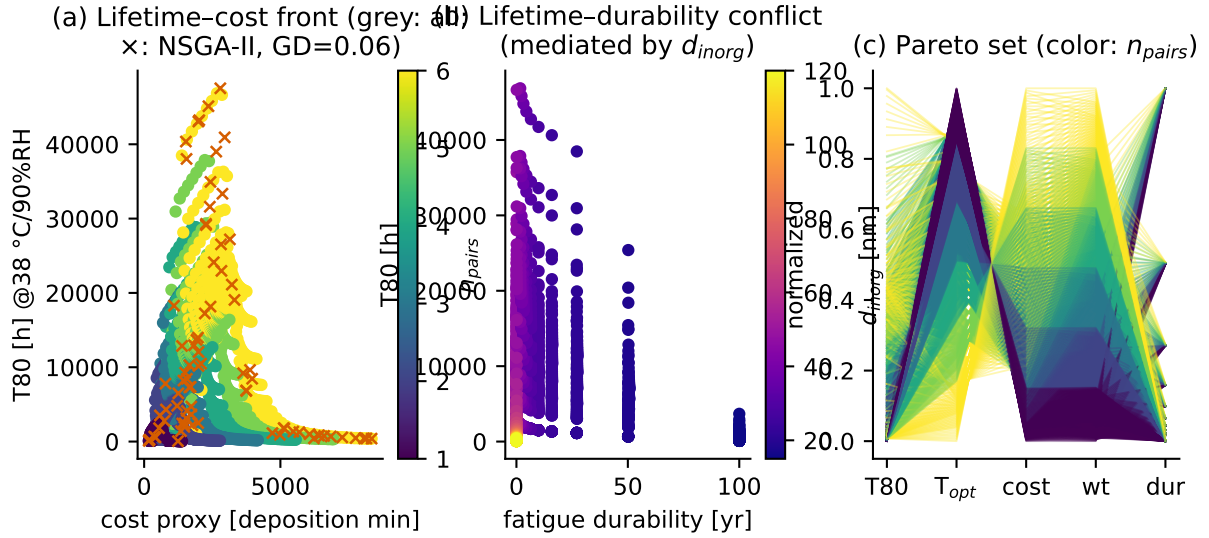


Figure 4: Six-objective Pareto exploration (4,104 designs). (a) Lifetime–cost front colored by dyad number; crosses: NSGA-II (GD = 0.06). (b) Lifetime–durability conflict mediated by d_{in} . (c) Pareto set in parallel coordinates.

Table 2: Representative Pareto designs (baseline parameters). Right column: moisture-limited lifetime under the device-anchored Tier-1 floor.

Design	t_{80}^{dose} [h] ^a	durability [yr]	cost [min]	mass [g/m ²]	floor lifetime
MAX 6 × (42/100)	46,857	1.7	2,671	1.53	~519 yr
KNEE 6 × (18/100)	25,399	50.2	1,202	1.10	~281 yr
ECONOMY 3 × (21/100)	15,983	27.1	693	0.58	~177 yr
single dyad 1 × (42/100)	2,356	—	445	—	~26 yr

^aAt the superseded placeholder $M_{\text{crit}}=0.01 \text{ g m}^{-2}$; retained for cross-figure consistency.

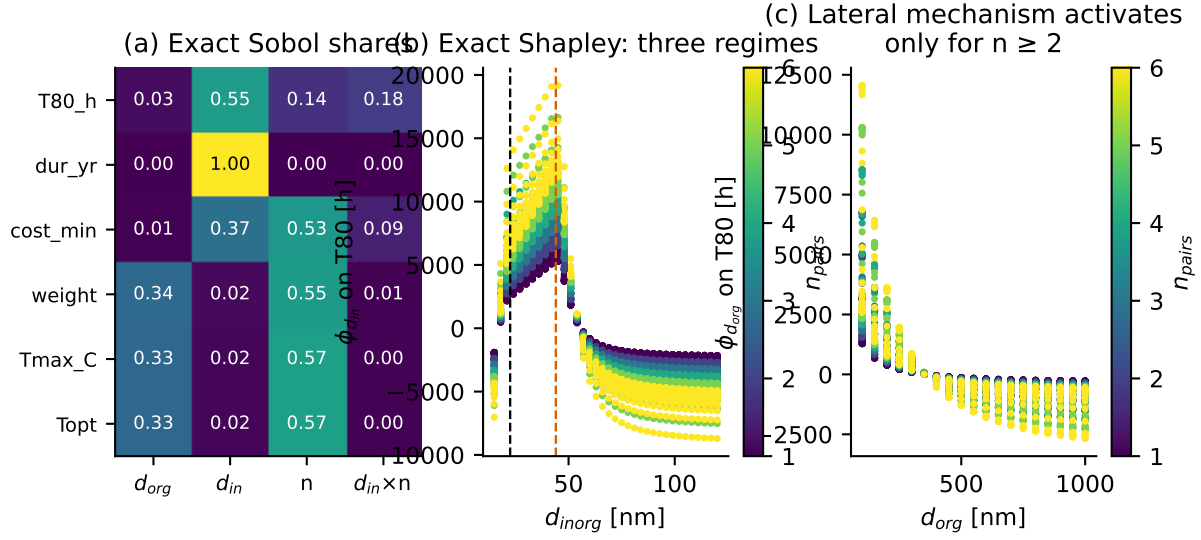


Figure 5: Exact explainability. (a) Sobol variance shares. (b) Exact Shapley dependence of lifetime on d_{in} with detected regime boundaries (22.5/44 nm). (c) The d_{org} effect activates only for $n \geq 2$ (number-verified; see Fig. 6).

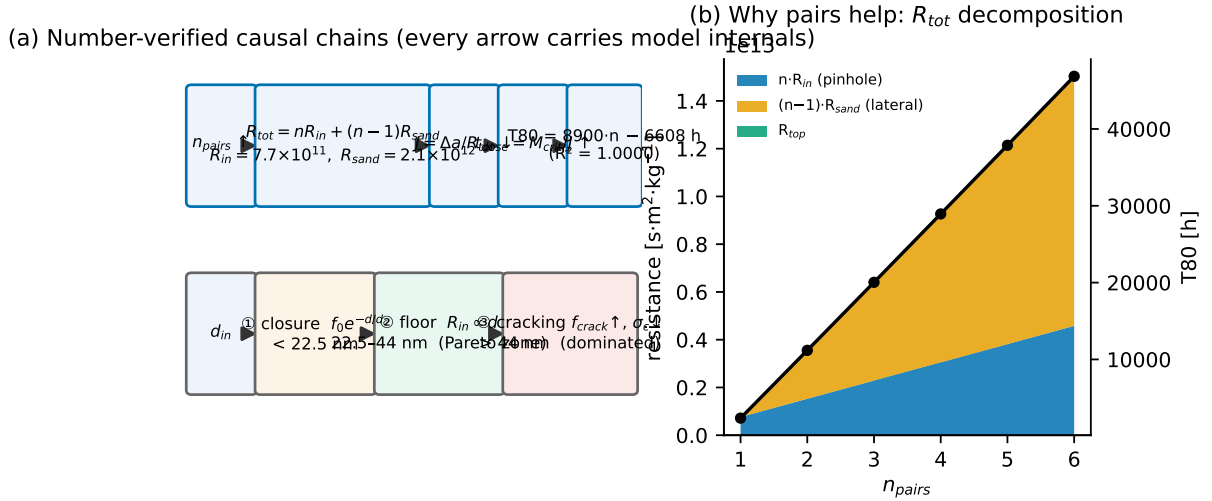


Figure 6: Number-verified causal chains. (a) Pair-count and inorganic-thickness chains with embedded model internals and regression checks. (b) Decomposition of R_{tot} versus dyad number.

Table 3: Design rules with confidence grades (full statements and validity conditions: SI Table S2).

Rule	Statement (abridged)	Confidence
G1	$d_{in} \geq 22.5$ [21.0–26.0] nm (pinhole closure)	HIGH
G2	$d_{in} < 44.0$ [37.5–48.0] nm (cracking; beyond: dominated)	HIGH
G3	$N_f \propto d^{-m/2}$; 10-yr needs $d_{in} \leq 30$ [16–49]; 16% infeasible	LOW-MED
G4	pair law linear (all draws); rel. gain 0.5 [0.4–0.5]/dyad	HIGH/MED
G5	organic: do not exceed planarization-required thickness	MED
G6	thermal/optical: not geometric degrees of freedom here	HIGH
G7	thresholds/rankings invariant to M_{crit} ; floored at 0.97 g m^{-2}	HIGH

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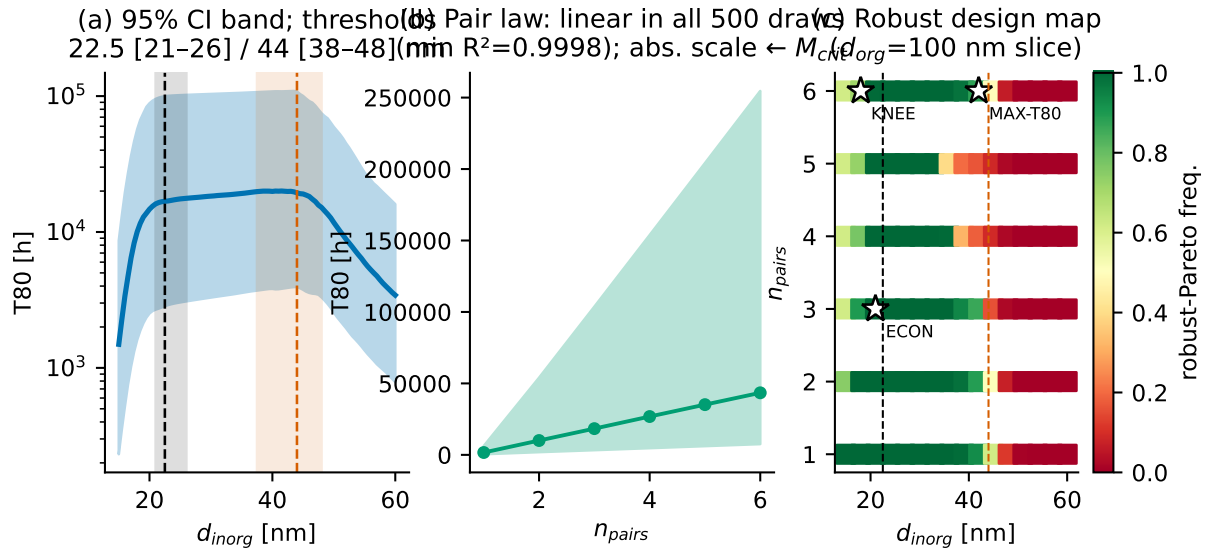


Figure 7: Robust guidelines from calibration-through Monte-Carlo ($N=500$). (a) Lifetime credible band with threshold CIs. (b) Pair law across all draws. (c) Robust-Pareto membership map with representative designs and thresholds.